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A Three-Stage Visual Training Evaluation Based on the METUX Model: Motivation, Engagement, and Well-Being

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ABSTRACT

This study presents a visual training design based on the Motivation, Engagement, and Thriving in User Experience (METUX) model to address low patient motivation and participation. It introduces a proactive approach by incorporating a playful scenario that alleviates digital eye strain and improves training outcomes. The study employs a three-phase experimental design: (1) validating the feasibility of visual training without additional equipment burden, (2) investigating the effects of gaze-controlled cursor movement on motivation and engagement, and (3) evaluating the impact of a playful training environment on well-being. Real-time eye-tracking is conducted using MediaPipe, with visual feedback provided through TouchDesigner, alongside data collection via physiological sensors. This research aims to transition visual training from passive to active, minimizing equipment complexity and enhancing user experience in vision regulation.

KEYWORDS

Proactive visual training;
METUX; progressive
experimental design;
motivation; engagement

1. Introduction

In the digital era, the widespread use of electronic screens and the increasing prevalence of online learning and remote work have led to digital eye strain becoming a significant and widespread health issue. The typical symptoms of digital eye strain include dry eyes, itching, a foreign body sensation, tearing, blurred vision, and headaches (Kaur et al., 2022). Research suggests that the prevalence of digital eye strain among computer and mobile device users may exceed 50%, particularly in individuals who engage in prolonged near-distance activities (Sheppard & Wolffsohn, 2018). This phenomenon has prompted extensive research into its causes, symptoms, and possible interventions. However, the effectiveness of current interventions remains limited by several factors.

Most existing studies focus on alleviating eye fatigue through screen time management, artificial tears, eye massaging devices, and specific eye exercises (Boopalan et al., 2025; Pardhan et al., 2022; Singh et al., 2021; M. T. M. Wang et al., 2019; Wolffsohn et al., 2023). However, these interventions present certain limitations. For instance, many eye massaging devices require users to invest in expensive hardware, while most eye relaxation exercises demand specific time and spatial constraints, limiting their practicality in everyday life. Additionally, while methods like Bates' training, based on eye movement exercises, have been shown to alleviate eye fatigue, these methods do not alter the refractive state of the eyes and lack personalized guidance for different user groups (Brata et al., 2023). More importantly, many existing approaches fail to sufficiently motivate user engagement, resulting in low adherence to eye relaxation exercises.

Nevertheless, vision remains one of the most relied-upon senses, directly influencing human interaction with the environment and others (Battaglini et al., 2022; Hutmacher, 2019; Ishikawa, 2023). Prolonged use of electronic screens often leads to significant discomfort, which not only affects work efficiency and academic performance but may also have adverse effects on quality of life and mental health (Kopilaš et al., 2025). Studies indicate that nearly half of individuals with vision impairments

report feeling disconnected from their surroundings and social interactions (Anthony & Kusnadi, 2021).

As such, effective interventions for digital eye strain are crucial, particularly in light of the increasing digitalization of daily life and remote work. Current research and interventions still exhibit significant gaps, particularly regarding convenience, device independence, and strategies to enhance user engagement. Therefore, there is an urgent need for an eye relaxation system that both alleviates eye fatigue and enhances user motivation and involvement.

This study aims to design and develop an innovative three-stage progressive eye relaxation training system. Grounded in evidence-based practices, the system emphasizes behavioral interventions and psychological motivation mechanisms, enabling users to perform eye relaxation exercises at any time, without the need for specialized equipment. This approach not only alleviates digital eye strain and improves visual comfort but also enhances users' proactive health awareness, contributing to their overall physical and mental well-being. The system will incorporate supplemental relaxation exercises derived from traditional Bates' training, such as gentle eye movements and alternating near and far focus, to enhance immediate comfort and visual awareness (Rawstron et al., 2005). However, it is important to note that these exercises will not alter the refractive state of the eyes but are primarily intended to alleviate fatigue and promote awareness of proactive relaxation (Svede et al., 2024).

Through the design and implementation of this system, this study will address existing gaps in research related to convenience, device independence, and personalized interventions, offering a scientifically effective and easily accessible solution to eye relaxation, thus improving eye health in the context of modern digitalized life.

2. Related work

2.1. Training methods and equipment requirements

In recent years, eye-tracking-based gamified interventions and multimodal visual training systems have demonstrated considerable potential in visual function rehabilitation. For example, in children with amblyopia, binocular gamified digital therapies utilize real-time eye-tracking to deliver precise binocular stimulation and visual feedback. This approach not only significantly improves visual acuity and stereopsis but also enhances user adherence and engagement (Yehezkel et al., 2022). Perceptual learning-based training, which integrates random dot patterns and action games, has been shown to foster neuroplasticity and improve visual sharpness and contrast sensitivity (Yeh et al., 2024). Additionally, virtual reality (VR) and augmented reality (AR) environments, with 3D stereoscopic stimuli, interactive scenes, and tactile and auditory feedback, offer effective accommodation training for conditions like amblyopia, strabismus, and myopia in children. Numerous studies have found that these approaches outperform traditional passive treatment methods (Chan et al., 2023; Saraiva et al., 2018; Tan et al., 2022).

However, these systems typically require specialized hardware (such as headsets or eye trackers), extended training durations, and often function as passive treatments in everyday use, which limits their accessibility and convenience. Furthermore, while most research and products focus on enhancing visual function, they often neglect users' psychological experiences, emotional regulation, and overall well-being. This oversight can create barriers to long-term adherence and active participation (Zanatta et al., 2022). Therefore, although hardware-dependent training methods effectively improve visual function, there is significant room for improvement in their accessibility, user-friendliness, and psychological support.

2.2. Theoretical framework and user engagement

While Flow Theory and Gamification models have advantages in enhancing user engagement, they do not systematically address the fundamental psychological needs of users, particularly regarding long-term intrinsic motivation and well-being. Flow Theory emphasizes deep immersion in activities but often lacks a thorough analysis of how different aspects of technology products can support or hinder

autonomy, competence, and relatedness (Bakker & van Woerkom, 2017). On the other hand, traditional gamification frameworks tend to focus on external incentives or game elements. These can enhance engagement in the short term but, if they fail to account for the internalization of users' basic needs, may lead to motivation becoming dependent on rewards, which can be unstable or result in negative rebound effect (Alberts et al., 2024; Deci et al., 1999; Hamari et al., 2014).

Moreover, when Flow Theory and Gamification models are applied, they often fail to assess the satisfaction of user needs across different contexts and stages of the user life cycle, limiting their applicability for designers looking for specific guidance for different stages (Oliveira et al., 2021; Xi & Hamari, 2019). As a result, these models are not entirely suitable for digital product design, especially systems centered on user needs.

2.3. Application of the METUX model

In contrast, the METUX model (developed by Peters et al., in 2018 and further applied by Jenö et al.) offers a more comprehensive and actionable framework. It systematically addresses the satisfaction of user needs at multiple levels (Jenö et al., 2021; Peters et al., 2018; Ryan & Deci, 2000; Sailer et al., 2017). Rooted in Self-Determination Theory (SDT), the METUX model helps designers evaluate how digital tools can foster autonomy, competence, and relatedness at various levels, such as the interface, task, behavior, and life domains, thereby enhancing users' intrinsic motivation, long-term engagement, and well-being (Burnell et al., 2023; Chaudhry et al., 2024), as illustrated in Figure 1.

At the adoption stage, METUX employs lightweight or device-free interaction designs that allow users to initiate eye relaxation training without additional hardware, reducing the barrier to entry and enhancing accessibility. The interface design offers clear feedback and flexible control, ensuring users feel in command of the training process, thereby fulfilling their autonomy and competence needs while providing supportive prompts to enhance relatedness. In the task domain, a three-stage progressive training approach ensures that each mini-exercise is independent and achievable, reinforcing the user's sense of competence and the meaningfulness of the tasks. Behaviorally, the system integrates relaxation training into daily life, encouraging long-term healthy behaviors. Personalized control mechanisms enhance adherence over time, promoting autonomy and relatedness while fostering positive health management behaviors. Finally, at the life level, users benefit from visible improvements in comfort and well-being, which enhance their overall life quality and demonstrate the positive impact of the technology on their personal lives.

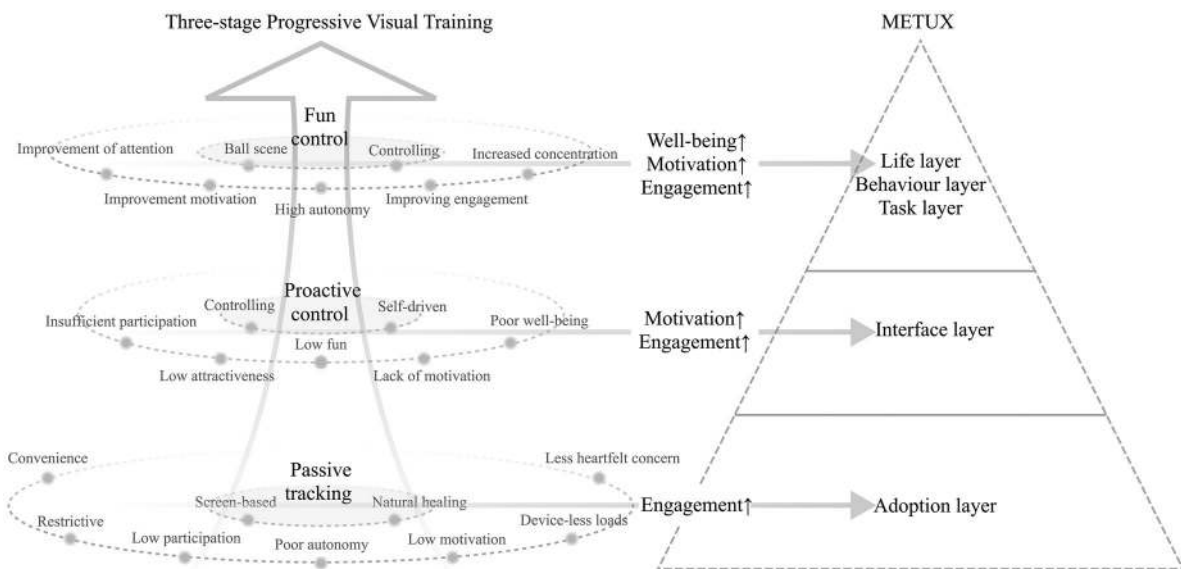


Figure 1. Three-stage visual training and its hierarchical mapping to the METUX framework: by constructing a three-stage visual training framework, this approach aims to progressively enhance user experience across the METUX multi-level architecture.

In summary, the METUX model's multi-level approach enables the design of low-barrier, highly autonomous, feedback-rich, and supportive user experiences, addressing the accessibility and psychological shortcomings of traditional, hardware-dependent passive visual training methods. It also drives sustained improvements in motivation, engagement, and well-being. Unlike Flow Theory and traditional gamification frameworks, METUX specifically emphasizes the ongoing fulfillment of users' basic psychological needs throughout long-term use, making it a robust framework for designing digital products centered on user motivation and well-being.

3. Design strategy

This system was designed from the outset with full consideration for ease of use across home and multi-scenario environments, enabling users to engage with it anytime and anywhere. It aims to seamlessly integrate ocular relaxation mechanisms into daily settings to help alleviate digital eye strain. In domestic contexts, the system emphasizes lightweight construction and user-friendly operation, ensuring that users can easily adopt it without professional guidance. This directly addresses the “interaction layer” of the METUX model. Such a design fosters strong usability and a sense of autonomous control during direct user–technology interaction, effectively bridging the gap left by conventional solutions that heavily depend on specialized equipment, lack user initiative, and often neglect psychological care. As a result, users can enjoy a comfortable and accessible experience at home or in any location, thereby alleviating eye fatigue more effectively.

Our design and validation process followed three progressive stages to explore how user motivation, engagement, and well-being could be continuously enhanced.

In the first stage, a slowly moving on-screen cursor guides users to follow it with their gaze, enabling eye movement exercises and brief relaxation without requiring any external hardware, thereby enhancing accessibility and initial engagement.

In the second stage, once users become familiar with the interface, a standard front-facing camera is employed for lightweight eye-tracking. Eye movements are then mapped to cursor navigation or interface interaction, reinforcing a sense of active control and deepening user engagement.

In the third stage, gamified scenarios are introduced on top of active interactions, using playfulness and positive emotional feedback to further encourage sustained participation.

Throughout the entire process, we adhere to a zero- or ultra-low-device-load interaction strategy, allowing users to engage with the system at any time without the need for additional hardware. Unlike conventional eye-tracking rehabilitation systems or VR/AR setups—which often emphasize visual function enhancement while overlooking the user's emotional experience—our system supports on-demand eye relaxation exercises while also attending to users' psychological well-being. The overarching goal is to substantially improve user engagement and long-term adherence, thereby promoting better eye health and overall quality of life in everyday contexts. The specific system framework is illustrated in Figure 2.

This study employs eye-tracking and physiological sensors to collect data and analyze users' behavioral and physiological characteristics. The goal is to validate whether the three-phase visual training, designed based on the METUX model, correlates with motivation and engagement, and whether the design possesses practical value and significance in enhancing users' well-being.

The fulfillment of autonomy and competence needs serves as a mediating factor between motivation and engagement (Ryan et al., 2006). When players are granted the ability to make choices within a game or participate in activities aligned with their interests and values, their perceived autonomy is strengthened (Rogers, 2017), subsequently leading to positive outcomes such as increased enjoyment.

The integration of active control with device-free interaction effectively reduces the bulkiness, discomfort, and interference typically associated with wearable devices (Amprimo et al., 2024), promoting innovation in traditional human–computer interaction (HCI) paradigms (Dezfuli et al., 2014). Current technological advancements in HCI—such as voice recognition, gesture and posture tracking, haptic feedback devices (Yin et al., 2020), eye-tracking, and biosensors (El Saddik et al., 2007; Gao et al., 2024)—enable the development of next-generation interaction systems. Specifically, eye-tracking

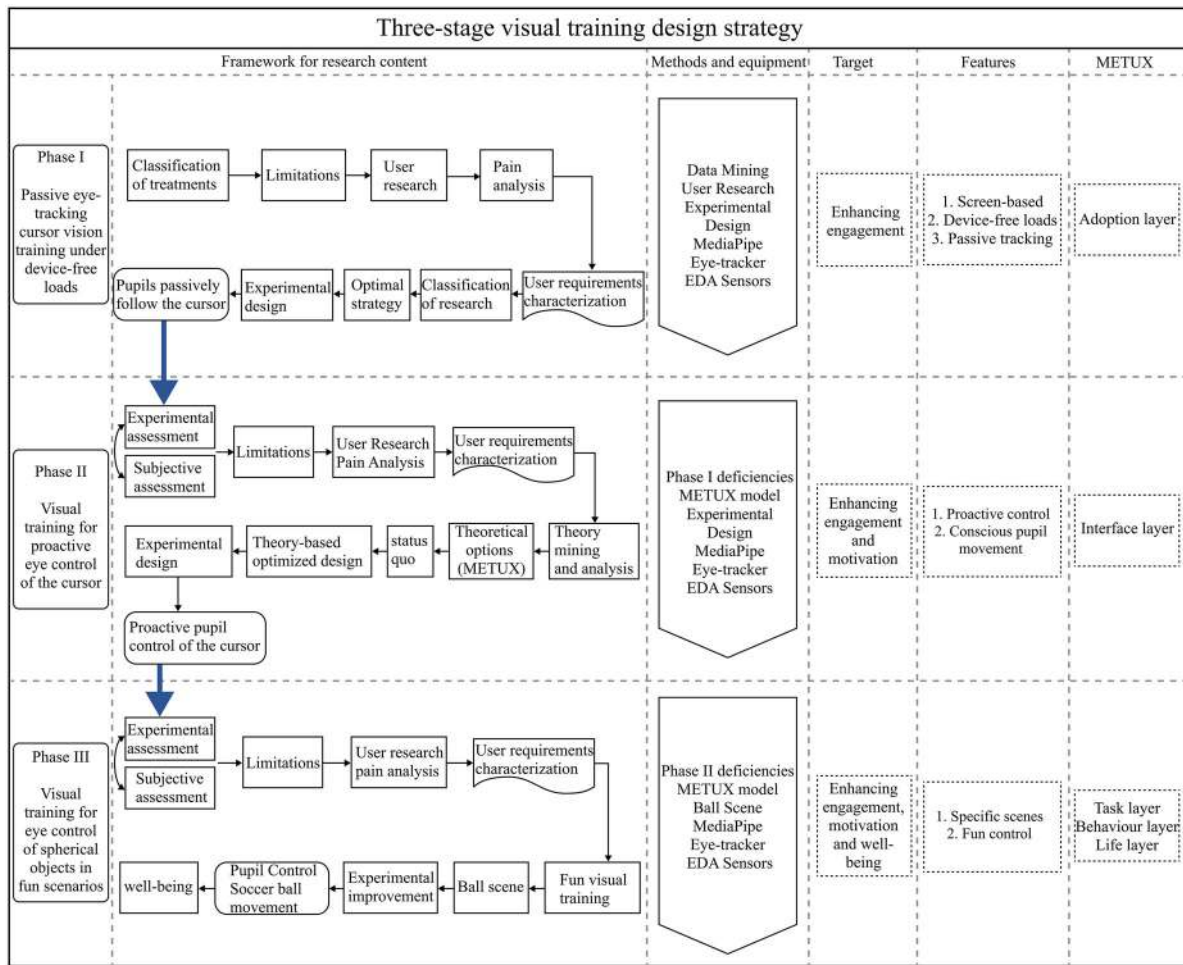


Figure 2. Three-stage visual training design strategy.

technology offers a novel device-free interaction method in digital screen contexts, helping to overcome the complexity and limitations of traditional contact-based interfaces (Klaib et al., 2021; Liao et al., 2023).

Conventional vision training usually involves tracking moving objects with the eyes, often grounded in methods such as the Bates technique or natural healing practices. These methods aim to improve the function of the ciliary muscles and crystalline lens, thereby slowing the elongation of the eyeball axis (Bates & Bates, 2008). However, such training tends to show delayed results and may suffer from reduced effectiveness due to low levels of motivation and engagement.

In response, this study incorporates eye-tracking technology based on the METUX model to enhance user motivation and engagement while eliminating the burden of physical devices. The design enables active interaction with digital screens via eye control, allowing users to achieve ocular relaxation and alleviate digital eye strain within an immersive experience.

For practical implementation, the study employs MediaPipe (MP) within TouchDesigner (TD) to realize eye-controlled cursor movement. This approach exercises the ocular muscles through voluntary control, aiming to relax the eyes and reduce fatigue. Given that ball-related activities have been shown to significantly improve dynamic visual acuity in young children (Jin et al., 2019; Zhou et al., 2023)—with large ball games demonstrating particularly strong effects—the study introduces a ball-based scenario in the third phase of training, as part of the METUX model framework. This addition enhances interactivity and autonomy, further supporting users' motivation, engagement, and sense of well-being.

The experiment aims to investigate the following key questions:

1. The effectiveness of device-free training and its potential impact on user motivation, engagement, and well-being;
2. Whether active visual training based on the METUX model leads to greater improvements in motivation, engagement, and well-being compared to passive training;
3. Whether the integration of a ball-based scenario within METUX-based active visual training further enhances users' motivation, engagement, and well-being.

To validate these hypotheses, the study analyzes behavioral and physiological data related to eye movement and bio-signals, exploring the changes in these parameters and their relationships with motivation, engagement, and well-being.

4. Materials and methods

4.1. Device description

This section focuses on how the system acquires eye-tracking data and electrodermal activity (EDA) signals. By integrating the MP plugin into the TD software platform, the eye tracker and the EDA sensor can synchronize data collection in real time. This enables the virtual environment to respond instantly to the user's eye movements while simultaneously recording physiological indicators, as illustrated in Figure 3. First, the hardware configurations and data acquisition workflows of the sensors are introduced. Then, the experimental design and key aspects of the virtual scene setup are explained. Next, the synchronization method between the eye tracker and the EDA sensor is described in detail. Finally, the preprocessing procedures for the collected physiological signals are outlined to ensure the accuracy and reliability of subsequent analyses.

4.1.1. MediaPipe

Google's MediaPipe Iris is a machine learning-based eye-tracking system that enables real-time tracking of the pupil and eye contours using a standard RGB camera, without the need for specialized hardware. This significantly reduces the cost and technical challenges typically associated with high-precision eye tracking, as shown in Figure 4. The system achieves a gaze estimation accuracy of 1.12 degrees in dynamic environments and 1.3 degrees in static conditions. The area of greatest stability is at the center of the screen, with an accuracy of approximately 2.53 degrees at the screen's edges. MediaPipe Iris also demonstrates excellent head movement compensation performance, adapting in real time to head

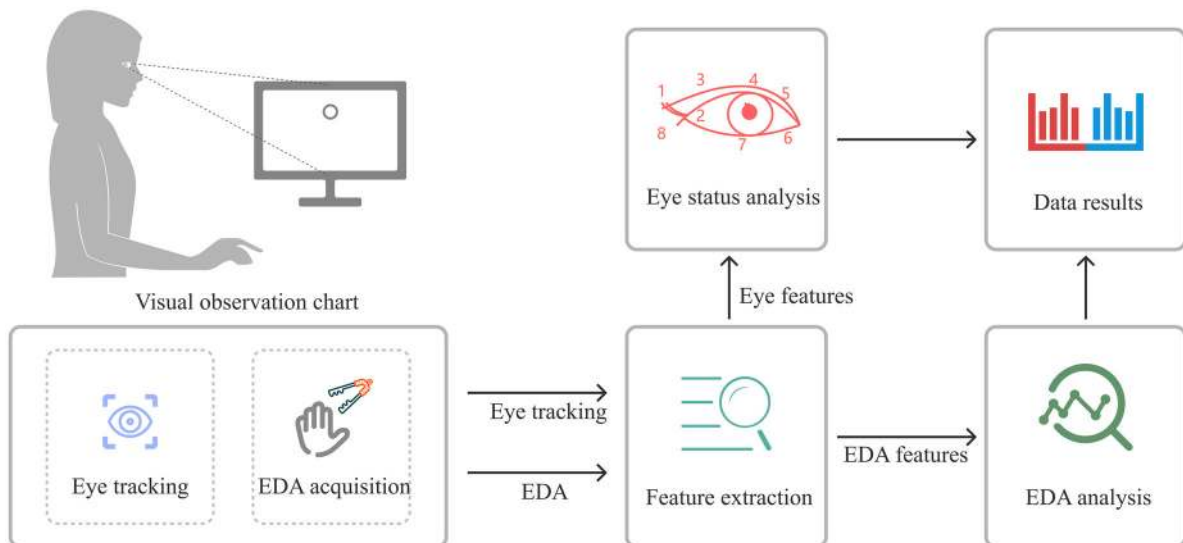


Figure 3. Experimental flow: following the arrows, proceed sequentially from eye-tracking data and EDA data acquisition through data analysis to the discussion of results.

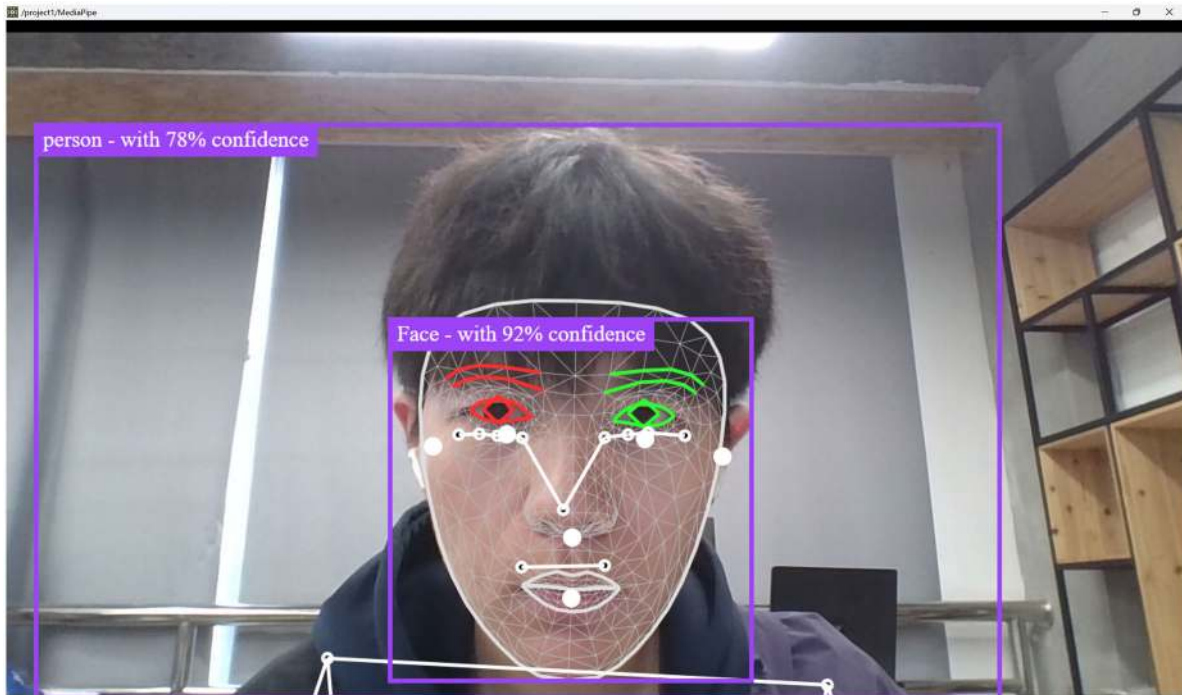


Figure 4. Using MediaPipe for eye capture.

displacements of up to 15%. Moreover, it maintains tracking stability through an ongoing calibration mechanism (Chhimpa et al., 2024).

Leveraging Google's MediaPipe framework, combined with GPU acceleration, the technology runs efficiently on both macOS and Windows platforms, supporting additional features such as facial recognition and gesture tracking. In this project, MediaPipe Iris is integrated into a system based on TD, which directly supports eye tracking. Utilizing TD's real-time data processing capabilities, the system efficiently processes eye movement data and provides real-time feedback. By optimizing data flow and event-driven mechanisms, the system maintains stable performance even in dynamic and interactive environments.

Integrating MediaPipe Iris with the TD platform offers an efficient, low-cost eye-tracking solution, reducing the need for specialized hardware.

4.1.2. Eye tracker

To accurately capture the participants' gaze behavior, this study employed the Tobii Pro Glasses 2 eye-tracking device, which uses near-infrared light patterns to determine gaze direction and supports wireless connectivity, thereby minimizing interference with the experimental process. Prior to the experiment, the device was calibrated and recording was controlled using the Tobii Pro Glasses Controller software. Once calibration was completed, the recorded video data was exported. The video and raw eye-tracking data were then imported into Ergolab 3.0 for preprocessing, including data cleaning and time synchronization. Eye movement metrics were subsequently calculated. Based on the pre-processed data, gaze metrics, fixation heatmaps, and eye movement path visualizations were generated to facilitate subsequent statistical analysis and data visualization.

4.1.3. EDA sensor

This study utilized the ErgoLAB system provided by Kingfar Technology for EDA analysis, with a sampling rate of 64 Hz. The system is compatible with both the human-machine environment synchronization platform and the physiological recording system, enabling offline processing and analysis of physiological signals related to EDA indicators. ErgoLAB 3.0 software supports a variety of filtering methods and skin conductance response (SCR) parameter extraction. It allows for precise extraction of

skin conductance level (SCL) and SCR indices, while also supporting cross-sequence analysis and advanced statistical analysis.

4.2. Experimental setup

This study proposes a novel approach that integrates MP techniques with TD mechanisms to capture eye movements and deliver visual input, enabling vision training without the need for conventional external devices. By actively regulating ocular motion, the method facilitates self-directed visual training, thereby reducing reliance on external equipment.

In the experiment, an eye tracker was used to record participants' ocular data in real time. Simultaneously, an EDA sensor was employed to measure sympathetic nervous system activation through sweat gland activity. Additionally, a subjective evaluation questionnaire was administered to assess participants' motivation and engagement levels, enabling a comprehensive quantitative analysis of the experimental data.

The experimental design included three groups:

1. Control Group: The on-screen cursor moved automatically while participants passively followed it with their eyes. The duration was 180 s.
2. Experimental Group A: The cursor remained stationary initially, and participants actively controlled its movement using their eye movements. The duration was 180 s.
3. Experimental Group B: Building on Experimental Group A, a ball interaction scenario was added. Participants controlled a football with their eye movements to move it in and out of a goal. The duration was 180 s (see Figure 5).

We collected eye-tracking metrics such as pupil diameter, blink frequency, and saccade patterns, along with physiological data from the EDA sensor across all three groups. A comparative analysis was conducted to explore the relationships among active eye-movement control, the motivational appeal of ball-related scenarios, engagement levels, and participants' sense of well-being.

4.2.1. Participants

This study was approved by the Ethics Committee of the School of Industrial Design at Hubei University of Technology and was conducted in the university's User Experience and Interaction Design Laboratory. To minimize environmental interference, the experiments were carried out in a noise-free setting, with a one-way mirror installed in the lab to facilitate unobtrusive observation of participants by the researchers, as illustrated in Figure 6.

A total of 20 participants were recruited, with ages ranging from 21 to 26 years (mean age = 23.5, SD = 1.1). Among them, 5 were female and 15 were male. Each participant completed three experimental conditions: a control group, Experimental Group A, and Experimental Group B. Each condition lasted 180 s. During the experiments, participants wore a Tobii eye tracker and an EDA sensor to collect physiological data. Due to limitations in time and resources, this study included 20 participants.

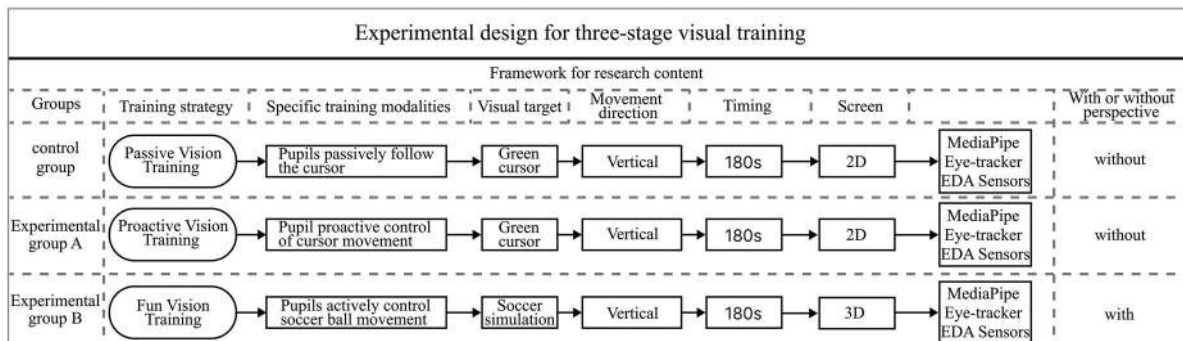


Figure 5. Experimental design.

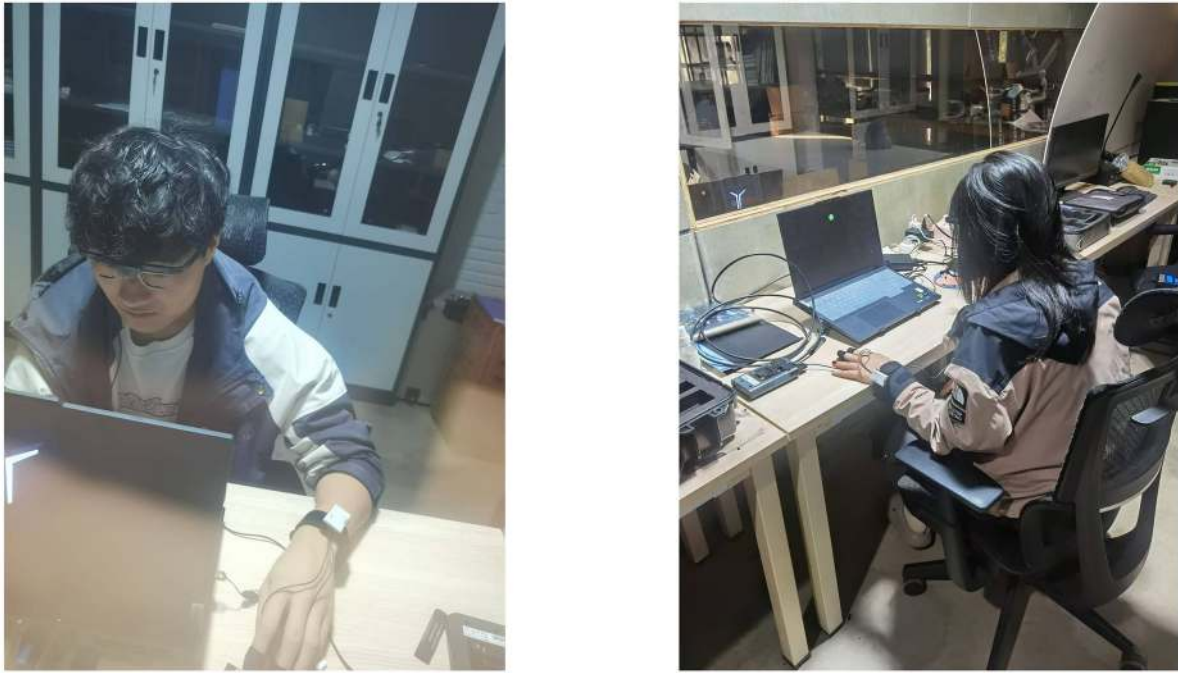


Figure 6. Experimenters and scenario equipment.

While the sample size is relatively small and the gender distribution is unbalanced, we consider the current sample size adequate for preliminary exploratory analysis, though it may limit the external validity of the results. Notably, the participants in this study are primarily from engineering disciplines, where the proportion of female students is relatively low, which partly accounts for the skewed gender distribution. Future research will aim to increase the sample size and achieve a more balanced gender distribution, thereby enhancing the statistical power and robustness of the findings.

In the data cleaning process, seven participants underwent a second round of experiments. For some participants, the EDA data showed irregularities, which remained unfit for analysis even after adjustments. Consequently, four sets of invalid EDA data were excluded. In total, 81 sets of eye-tracking data and 69 sets of valid EDA data were collected, resulting in 150 valid experimental data sets.

Before the experiment, participants viewed a resting screen for 5 s and underwent a pupil control adaptation session designed for the experimental group. This training enabled them to independently adjust their sitting posture and viewing angle without operator assistance, ensuring smooth completion of the experiment. After the experiment, participants were asked to complete a questionnaire to support the evaluation of their experience and the experimental outcomes.

4.2.2. Tasks and stimuli

The experimental visual tasks were categorized into three groups: the control group performed an eye-tracking cursor task (Figure 7a), Experimental Group A controlled vertical cursor movement using eye gaze (Figure 7b), and Experimental Group B directed a football into a goal using eye gaze (Figure 7c,d).

Considering that the sensitivity of vertical eye movement is only about half that of horizontal movement, with a more limited range of motion (Ohlendorf et al., 2022) and lower frequency, the tasks were primarily designed to involve vertical movement on the screen. This was intended to more effectively exercise the eye muscles. The cursor was rendered in a green hue, which is generally more comfortable for the eyes (van den Berg et al., 2015). Leveraging the phenomena of visual persistence and critical flicker fusion frequency (Figure 8) (Duchowski, 2017), the cursor was programmed to move passively every 0.8 s to facilitate eye-tracking engagement.

The cursor moves vertically on the screen at a frequency of 1.25 cycles per second, with a cycle duration of 8 s. Its movement spans from the top to the bottom of the screen, ensuring that the range is large enough to fulfill the task requirements.

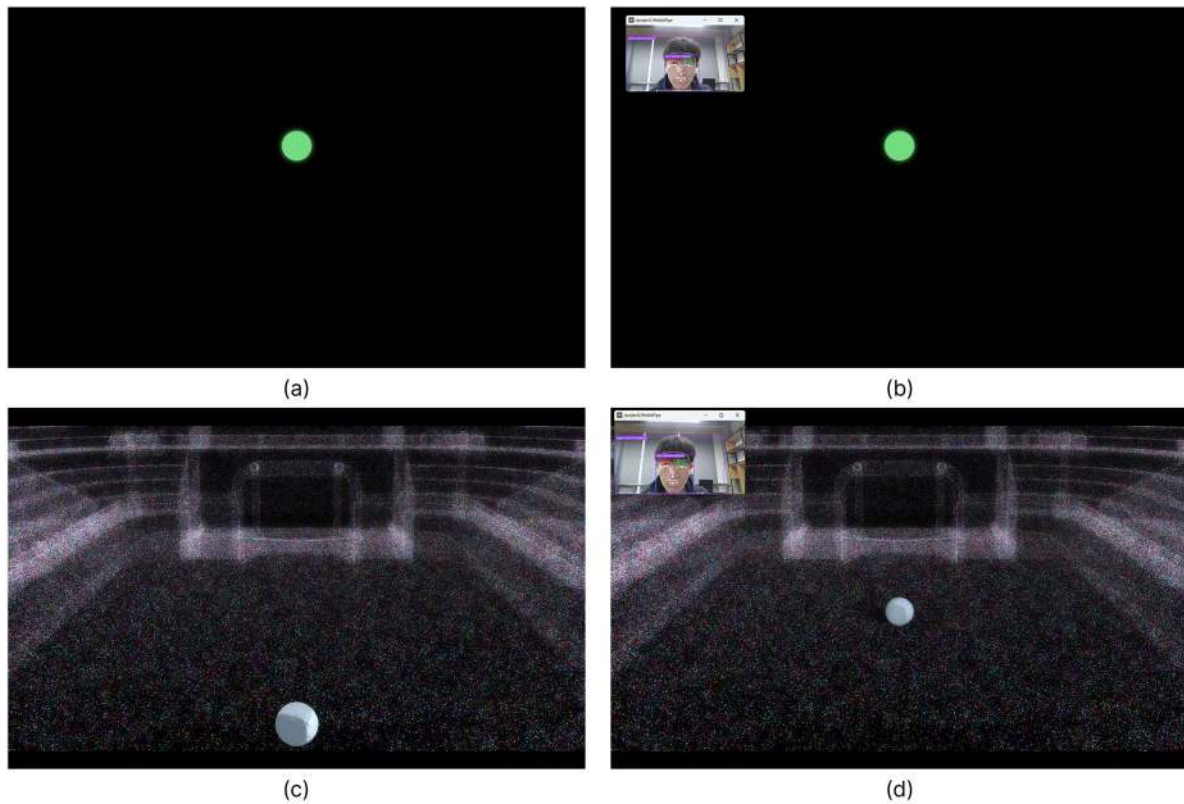


Figure 7. Control visual stimulus (a), experimental group a visual stimulus (b), experimental group B visual stimulus (c, d).

In experimental group B, the football continuously follows the participant's point of gaze. Specifically, the participant's gaze direction determines both the movement direction and position of the football, which tracks the changes in the gaze direction in real-time until it reaches the target area.

4.2.3. Environmental control

To ensure the accuracy and reproducibility of the experimental data, it is essential to rigorously control key environmental variables such as screen distance, brightness, and viewing posture. The experiment was conducted using a 2023 Lenovo Y9000P laptop (16 inches, 2560×1600 resolution, 500 nits brightness). The distance from the screen to the participant's eyes was uniformly controlled at 60 cm, which aligns with the ergonomically recommended viewing distance. A tape measure was used for calibration to maintain consistency. The screen brightness was set to 200 nits, which effectively minimizes visual fatigue under typical indoor lighting conditions (300–500 lux), avoiding both overly dim and excessively bright conditions. As for viewing posture, participants were instructed to sit upright, with their backs firmly against the chair's backrest, and their eyes should be aligned with the upper third of the screen's height. The screen angle was maintained between 110° and 120° to ensure the optimal viewing angle for the laptop. During the experiment, on-site personnel supervised the participants to ensure consistent control of the environmental variables.

4.2.4. Procedure

Before the formal experiment began, participants were required to first view a resting screen for 5 s, allowing them to relax and stabilize their EDA and eye movements. This phase, known as the baseline period, refers to the time before the stimulus presentation when no other tasks are required aside from focusing on the screen.

Following this, the experimental group underwent eye control training to ensure they could independently complete the entire experiment without operator intervention. This phase, called the task

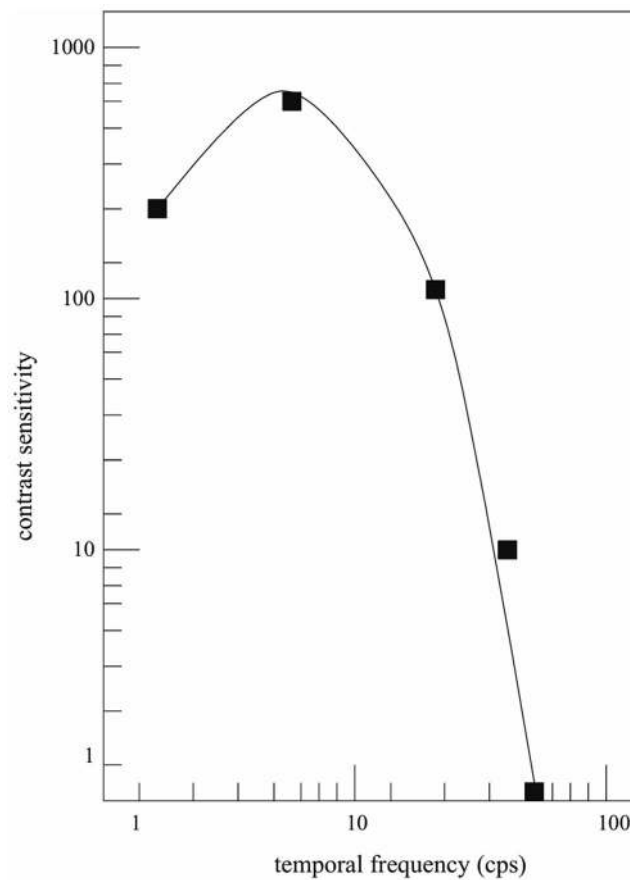


Figure 8. Critical fusion frequency.

stimulation period (C.-A. Wang et al., 2018), aimed to familiarize participants with the task content and operational procedures.

Once the formal experiment began, participants were required to experiment in a private space, ensuring no external interference from others, and minimizing noise disturbances as much as possible. The laboratory was equipped with a one-way mirror to facilitate monitoring by the observer.

After completing preparatory tasks (such as reading instructions, signing the informed consent form, wearing the EDA sensor, performing impedance checks, and calibrating the eye tracker) and taking a short break between experimental blocks, each participant was expected to complete three sets of experimental tasks within 12 to 20 min. Participants were allowed to take breaks or withdraw from the experiment for personal reasons during the process. Afterward, participants were asked to complete a questionnaire to provide subjective evaluation data.

To control for the potential order effects in the experiment, this study adopted a randomized order of presentation to balance the influence of the sequence in which different experimental conditions were presented on participants' performance. Since a within-subjects design was used, each participant underwent all experimental conditions (control group, experimental group A, experimental group B). In order to reduce any bias due to the sequence of conditions, the order of presentation was randomized for each participant. This approach ensured that the sequence of experimental and control conditions was varied, minimizing any differences in results that might arise from a specific order. Furthermore, to mitigate the impact of order effects, appropriate rest periods were incorporated between tasks. These breaks between experimental conditions allowed participants to recover and avoid fatigue, which could otherwise interfere with the results. Additionally, the design of the experimental tasks accounted for differences in task difficulty, ensuring that more challenging tasks did not disproportionately affect the performance in subsequent conditions.

4.2.5. Data extraction and processing

To remove artifacts and noise from the data, the processing procedure is divided into three steps:

Step 1: Apply filters to eliminate signal noise and artifacts. The ErgoLAB 3.0 system offers several filtering methods, including sliding mean filtering, Gaussian filtering, Hann filtering, and low-pass filtering. This study primarily employs Gaussian and sliding median filtering techniques to enhance signal quality.

Step 2: Conduct manual signal correction. ErgoLAB 3.0 provides multiple manual correction options, including linear interpolation, spline interpolation, and area-based signal copying. These methods are designed to repair or exclude segments of the signal that show clear anomalies.

Step 3: Extract eye movement video data and EDA signal data. For each experimental group, a full three-minute video clip is extracted. Simultaneously, EDA signals are processed by setting a custom window length and SCR threshold to identify and remove artifacts linked to specific events, ensuring each dataset retains uninterrupted, complete three-minute data. To further enhance data quality, the system includes time-domain analysis capabilities, automatically computing statistical metrics such as the mean, maximum, minimum, standard deviation (SD), and variance of the EDA signals. This helps in identifying and excluding outliers.

During the statistical analysis phase, descriptive statistics for each subject are presented as “mean $\pm$ standard deviation,” based on whether the data follow a normal distribution. SPSS software was used for independent sample t-tests, analysis of variance (ANOVA), and correlation analysis to compare behavioral and physiological indicators among the three experimental groups. The analysis aims to determine whether significant differences exist between the various indicators during eye movement, which could indicate their validity as measures of motivation or engagement.

Both t-tests and ANOVA yield a probability value (p), which quantifies the difference between the data distributions. Following commonly accepted standards in the literature, when $p \leq 0.05$, the results are deemed statistically significant, indicating a meaningful difference between the two groups of data. Furthermore, paired sample comparisons provide a clearer understanding of the specific differences between data characteristics, offering stronger empirical evidence.

5. Experimental results

This section presents the analysis of the eye movement and EDA data, providing a comprehensive evaluation of the differences across the three experimental groups. It further compares the participants' performance in terms of motivation, engagement, and well-being.

The error bars in the figures represent the standard error (SE), which quantifies the variability in the estimate of the sample mean. A larger standard error indicates greater uncertainty in the estimate, suggesting lower precision in the sample mean. All error bars are derived from the sample data of each experimental condition, and they are used to visualize the differences between groups, as well as to indicate the statistical significance of these differences.

5.1. Behavioral results

In this study, the average of 27 sets of eye movement data from each group was calculated for statistical comparison, with results reported to three decimal places. As shown in [Figure 9](#), the mean and standard deviation for the pupil diameters and their differences between the maximum and minimum values are as follows: For the control group, the mean pupil diameter was 4.565 mm (SD = 0.618), the minimum pupil diameter was 3.236 mm (SD = 0.490), and the maximum pupil diameter was 5.574 mm (SD = 0.560), resulting in a pupil diameter difference of 2.337 mm (SD = 0.463). For Group A, the mean pupil diameter was 4.418 mm (SD = 0.583), the minimum pupil diameter was 3.229 mm (SD = 0.525), and the maximum pupil diameter was 5.484 mm (SD = 0.608), with a pupil diameter difference of 2.255 mm (SD = 0.555). For Group B, the mean pupil diameter was 4.082 mm (SD = 0.495), the minimum pupil diameter was 2.935 mm (SD = 0.398), and the maximum pupil diameter was 5.193 mm

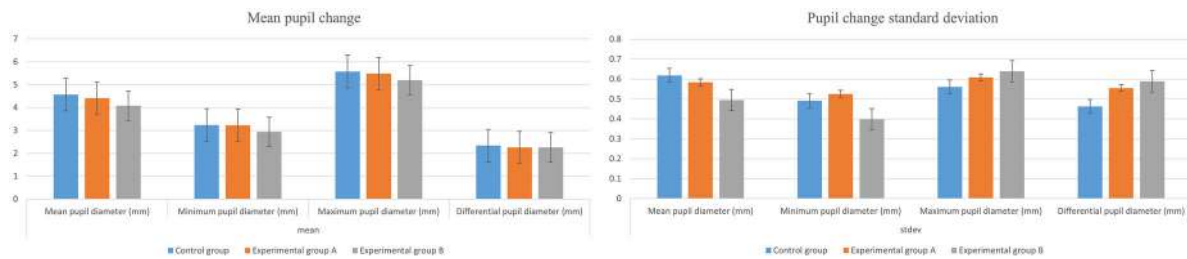


Figure 9. Analysis of pupil size change.

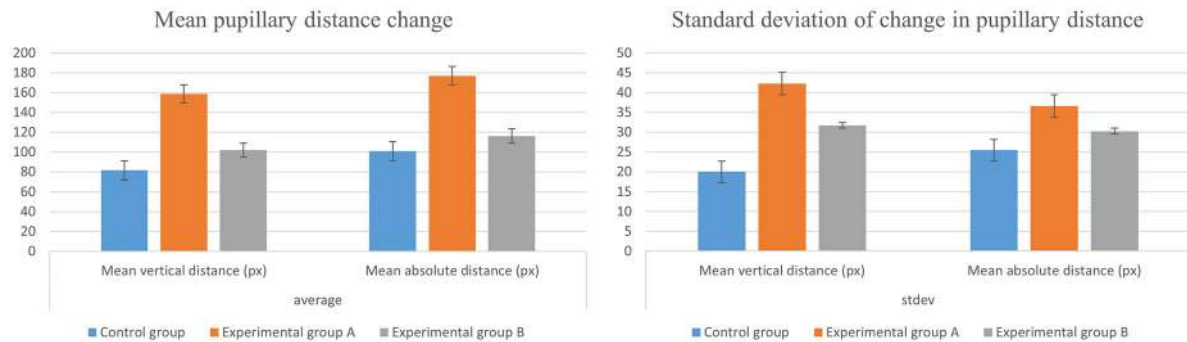


Figure 10. Analysis of pupillary distance variation.

(SD = 0.639), resulting in a pupil diameter difference of 2.258 mm (SD = 0.589). These descriptive statistics serve as the foundation for subsequent group comparisons.

According to the experimental design, comparisons were made as follows: first, between the control group and experimental group A, then between experimental group A and experimental group B, and finally, a one-way ANOVA was performed on all three groups. The independent samples t-test between the control group and experimental group A showed no statistically significant differences: average pupil diameter ($p = 0.375$), minimum pupil diameter ($p = 0.960$), and maximum pupil diameter ($p = 0.570$). The independent samples t-test between experimental groups A and B revealed significant differences in both the average pupil diameter ($p = 0.027$) and minimum pupil diameter ($p = 0.024$), while the maximum pupil diameter ($p = 0.093$) was close to the threshold for significance but did not reach conventional levels. In the comparison between the control group and experimental group B, all pupil diameter measurements were significantly different: average pupil diameter ($p = 0.003$), minimum pupil diameter ($p = 0.017$), and maximum pupil diameter ($p = 0.024$). The results of the overall one-way ANOVA across all three groups indicated significant differences in the average pupil diameter ($F(2, 78) = 5.118$, $p = 0.008$, Partial $\eta^2 = 0.116$) and minimum pupil diameter ($F(2, 78) = 3.541$, $p = 0.034$, Partial $\eta^2 = 0.083$), while the maximum pupil diameter ($F(2, 78) = 2.934$, $p = 0.059$, Partial $\eta^2 = 0.07$) was close to significant but typically considered a borderline result. These findings suggest that: experimental groups A and B showed significant changes in both average and minimum pupil diameters; the control group and experimental group B exhibited significant differences in all pupil diameter measures; and the three groups overall showed significant differences in average and minimum pupil diameters, with the maximum pupil diameter difference approaching significance. Further interpretation should take into account effect sizes, assumption checks, and adjustments for multiple comparisons.

The experimental results show that changes in pupil size are associated with variations in the vertical and absolute distances of the fixation points. As illustrated in Figure 10, the study compared the mean vertical distance (MVD) and the mean absolute distance (MAD) across the three groups. Both experimental groups A and B exhibited significantly higher MVD and MAD values compared to the control group (ANOVA: MVD $F(2, 78) = 40.317$, $p < 0.001$, Partial $\eta^2 = 0.508$; MAD $F(2, 78) = 45.346$, $p < 0.001$, Partial $\eta^2 = 0.538$). To further assess eye movement distribution under different conditions, 27 eye trajectories from each group were extracted, and heatmaps were generated to analyze the

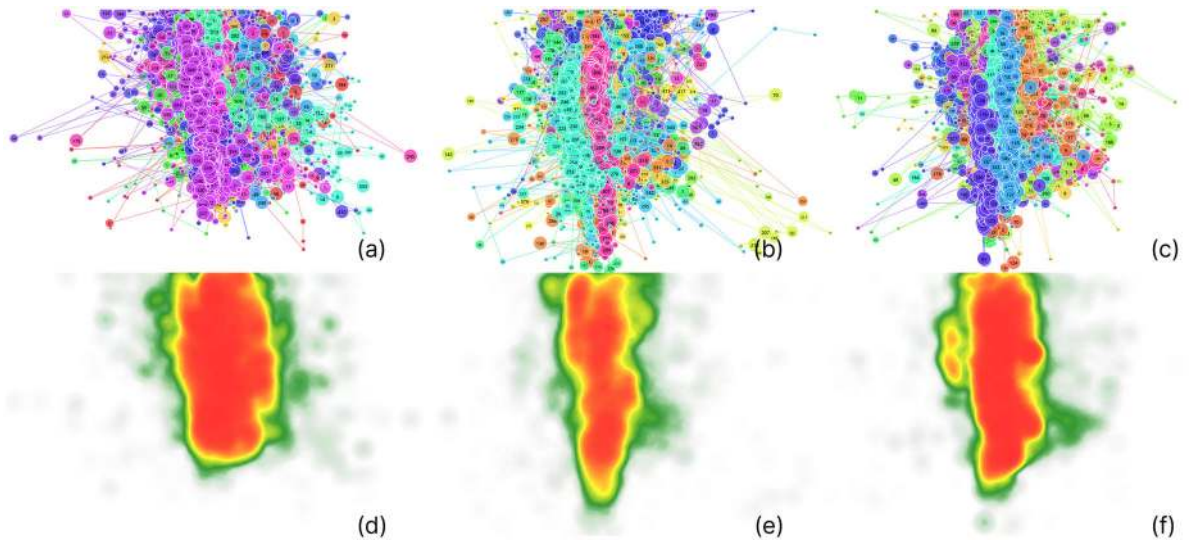


Figure 11. Thermograms of pupil trajectories, from left to right: control group (a, d), experimental group A (b, e), experimental group B (c, f). Pupil trajectories from different groups were captured for comparative analysis of focus levels.

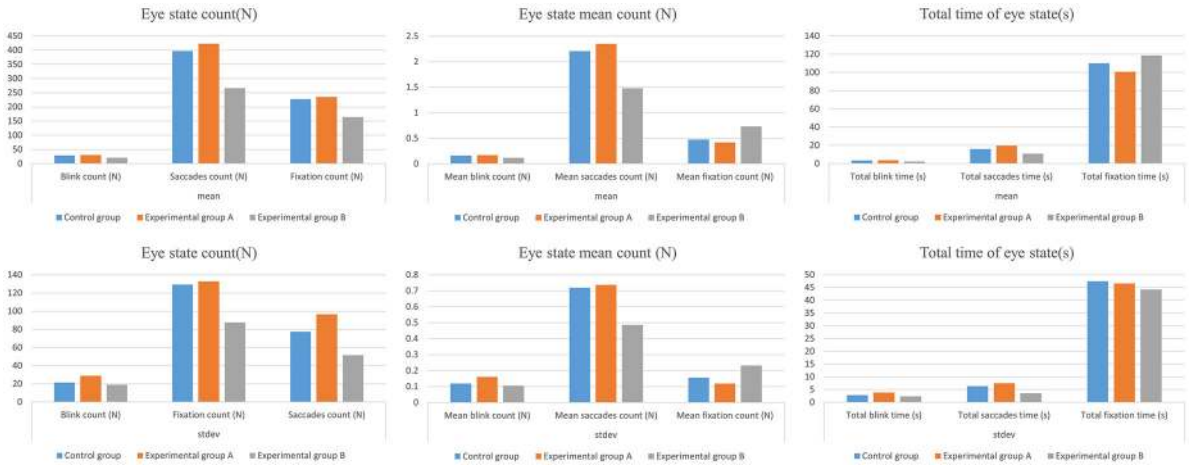


Figure 12. Eye state analysis: Number of blinks, eye twitches, and fixations; average frequency and total duration.

patterns and randomness of these trajectories (See Figure 11: Control group (a) (d), Experimental group A (b) (e), Experimental group B (c) (f)). These analyses aid in understanding the impact of visual training on fixation distribution and stability and provide a foundation for discussing the underlying mechanisms of the experiment.

This study further explores the relationship between eye state indicators and motivation and focus, as illustrated in Figure 12. Regarding blinking, the blink count, average blink count, and total blink duration in Experimental Group B were all lower compared to those in the control group and Experimental Group A. However, the results from the one-way ANOVA (blink count $F(2, 78) = 1.202$, $p = 0.306$, Partial $\eta^2 = 0.03$; average blink count $F(2, 78) = 1.202$, $p = 0.306$, Partial $\eta^2 = 0.03$; total blink duration $F(2, 78) = 1.311$, $p = 0.275$, Partial $\eta^2 = 0.033$) and the independent sample t-tests between the control group and Experimental Group B (blink count $p = 0.167$, average blink count $p = 0.167$, total blink duration $p = 0.153$) showed no significant differences. Correlation analysis indicated varying relationships between blink count and MVD across the groups: in the control group, $r = -0.247$ (near no correlation), in Experimental Group A, $r = -0.538$ (moderate correlation), and Experimental Group B, $r = -0.370$ (weak correlation). These findings suggest the need for future research to further explore the potential relationship between blinking and gaze distance, or other

related indicators, in order to better understand how active eye movement control may influence blinking behavior.

Regarding saccades, Experimental Group B exhibited significantly fewer saccades and shorter corresponding durations compared to the control group and Experimental Group A. Both one-way ANOVA (saccades count $F(2, 78) = 13.635$, Partial $\eta^2 = 0.259$; average saccades count $F(2, 78) = 13.635$, Partial $\eta^2 = 0.259$; total saccades duration $F(2, 78) = 14.819$, Partial $\eta^2 = 0.275$) and independent sample t-tests showed $p < 0.001$, suggesting that gamified active training could potentially reduce the frequency and duration of saccadic shifts. Concerning gaze, while an Area of Interest (AOI) was not defined, engagement was assessed based on total gaze duration and frequency. Experimental Group B showed a higher total gaze duration but fewer gazes compared to the other groups. The one-way ANOVA revealed significant differences for gaze count ($F(2, 78) = 6.891$, $p = 0.002$, Partial $\eta^2 = 0.15$), total gaze duration ($F(2, 78) = 1.029$, $p = 0.362$, Partial $\eta^2 = 0.026$), and average gaze duration ($F(2, 78) = 24.419$, $p < 0.001$, Partial $\eta^2 = 0.385$), suggesting that Experimental Group B exhibited a more stable gaze pattern. A combined analysis of blinking and gaze duration provides insights into the usability of blink data and its connection with motivation and engagement. This section requires further exploration of the mechanisms and implications in the discussion.

5.2. Physiological results

The analysis of EDA signals in this study was carried out in two stages: preprocessing and feature extraction. Preprocessing techniques, including Gaussian filtering, detrending, and interpolation, were applied to reduce noise and artifacts, thereby enhancing signal quality. Feature extraction was then performed in the time domain to compute various statistical measures for skin conductance (SC), SCL (also referred to as tonic), and SCR (also referred to as phasic), as illustrated in Figure 13.

The mean SC values for each group were as follows: control group $5.74 \mu S$ (SD = 0.40), experimental group A $5.88 \mu S$ (SD = 0.36), and experimental group B $6.06 \mu S$ (SD = 0.46). The mean SCL values for each group were: control group $5.52 \mu S$ (SD = 0.35), experimental group A $5.68 \mu S$ (SD = 0.28), and experimental group B $5.86 \mu S$ (SD = 0.37). The mean SCR values for each group were: control group $0.22 \mu S$ (SD = 0.22), experimental group A $0.20 \mu S$ (SD = 0.16), and experimental group B $0.20 \mu S$ (SD = 0.17).

The trends observed in SC and SCL were as follows: experimental group B > experimental group A > control group, while SCR data indicated: control group > experimental group A = experimental group B (see Figure 13). No significant differences were found between the three groups based on the current sample size and analysis methods. Further research involving a larger sample or different experimental conditions may be required to confirm these results.

5.3. Subjective evaluation: System effectiveness and visual fatigue relief

5.3.1. Satisfaction

Following the completion of the experiment, questionnaires were designed and administered to the three visual training groups, as outlined in the research plan. These questionnaires were used to assess and quantify the participants' performance and their interactive experience throughout the training process, to evaluate the effectiveness of the three-stage visual training strategy. The reliability of the questionnaire data was then analyzed using SPSS, yielding an overall Cronbach's $\alpha = 0.888$, indicating good

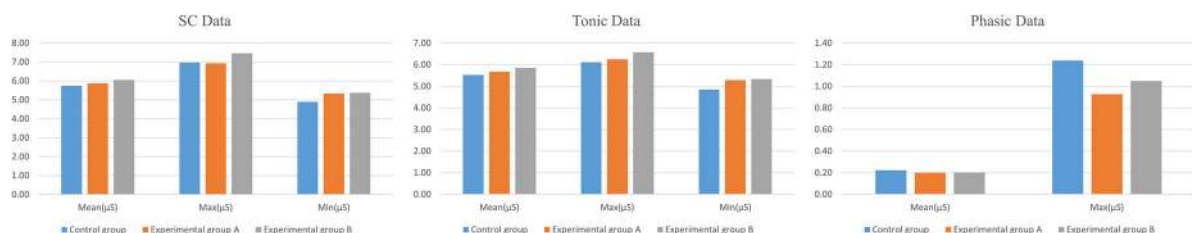


Figure 13. EDA data: SC data, SCL(tonic) data, SCR(phasic) data.

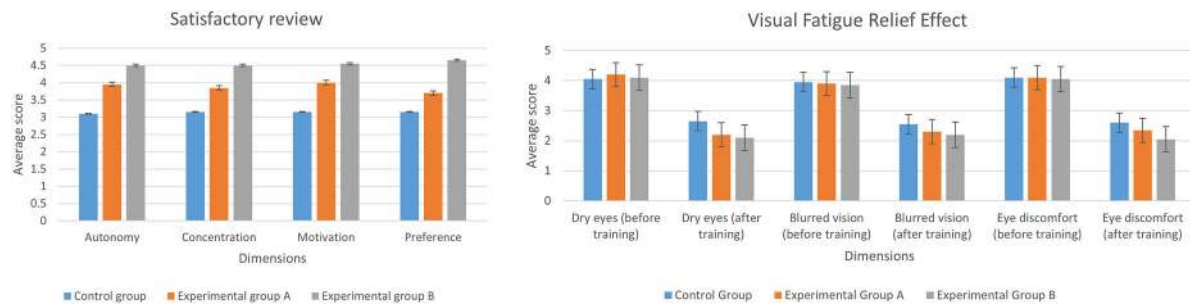


Figure 14. Subjective evaluation: Satisfaction review and visual fatigue relief effect.

internal consistency of the scale. The ANOVA results showed that, in terms of autonomy, concentration, motivation, and preference, Experimental group B > Experimental group A > Control group. The differences observed were statistically significant ($p < 0.001$), as illustrated in Figure 14.

5.3.2. Visual fatigue relief effect

To assess the effectiveness of the system in relieving visual fatigue, this study employed self-reported questionnaires addressing three aspects of visual fatigue: eye dryness, blurred vision, and eye discomfort. A 1–5 scale was used for each dimension, with 1 indicating no symptoms and 5 indicating very severe symptoms. A total of 20 participants took part in the study, which was divided into a Control Group, Experiment Group A, and Experiment Group B, based on different training modes. Scores were recorded before and after the training, and the data were analyzed, as illustrated in Figure 14.

Before the training, all three groups exhibited significant visual fatigue. Specifically, the Control Group reported scores of 4.05 for eye dryness, 3.95 for blurred vision, and 4.1 for discomfort; Experiment Group A reported scores of 4.2, 3.9, and 4.1, respectively; and Experiment Group B reported scores of 4.1, 3.85, and 4.05. These results indicate that all three groups experienced substantial visual fatigue before the training. One-way ANOVA showed no significant differences in these scores between the groups (eye dryness: $p = 0.517$, blurred vision: $p = 0.831$, discomfort: $p = 0.962$). Main effect analysis revealed $F(2, 57) = 0.668$ (Partial $\eta^2 = 0.023$) for eye dryness, $F(2, 57) = 0.186$ (Partial $\eta^2 = 0.06$) for blurred vision, and $F(2, 57) = 0.039$ (Partial $\eta^2 = 0.001$) for eye discomfort, suggesting minimal differences in visual fatigue across the groups before the training.

Post-training, all groups reported notable improvements in visual fatigue relief. Independent sample t-tests showed significant differences in the reduction of fatigue across the groups ($p < 0.001$). Specifically, Experiment Group B's scores dropped to 2.1 for eye dryness, 2.2 for blurred vision, and 2.05 for discomfort, demonstrating the most pronounced relief; Experiment Group A's scores decreased to 2.2, 2.3, and 2.35, showing a good level of relief; whereas the Control Group's scores fell to 2.65, 2.55, and 2.6, reflecting some improvement, though the relief was less significant. One-way ANOVA results (eye dryness: $p = 0.003$, blurred vision: $p = 0.171$, discomfort: $p = 0.055$) further demonstrated that the improvement in eye dryness was the most significant ($F = 6.293$, Partial $\eta^2 = 0.181$), while the changes in blurred vision ($F = 1.821$, Partial $\eta^2 = 0.06$) and eye discomfort ($F = 3.055$, Partial $\eta^2 = 0.097$) were less pronounced.

In conclusion, Experiment Group B showed the most significant effect in relieving visual fatigue, followed by Experiment Group A, with the Control Group showing the least improvement.

6. Discussion

This section analyzes and discusses the research findings, including their significance, comparison with existing research, and the limitations of the study. First, we examine the feasibility of transferring passive training to electronic screens, particularly the possibility of conducting training without the need for additional equipment. Second, based on this feasibility, we compare the data of active and passive training, investigating whether active training can enhance user motivation and engagement. Finally, we evaluate the impact of incorporating gamified design elements into active training, analyzing whether this can more effectively enhance user motivation, engagement, and well-being. Table 1 illustrates the

Table 1. Mapping of behavioral and physiological indicators to METUX. Arrows ↑ and ↓ denote increase/enlargement and decrease/reduction, respectively.

METUX	Behavioral indicators	Physiological indicators	Supporting literature
Motivation	Pupil dilation↑ Gaze duration↑	SCR amplitude↑ SCL relative↑	Le et al. (2019); Schneider et al. (2018) Budnik-Przybylska et al. (2024); Lou et al. (2019)
Engagement	Gaze duration↑ Blink rate↓ Frequency of eye-beating/path focusing↑	SCR frequency↑ SCL fluctuation↑	Negi and Mitra (2020); Zhang et al. (2012) Di Lascio et al. (2018); Ranti et al. (2020) Bonneh et al. (2011); Lai et al. (2013)
Well-being	Blink rate↓ Pupil baseline stabilization	SCL relative ↑ SCR frequency ↑	Jang et al. (2019); Maffei and Angrilli (2019) Khalfa et al. (2002); McCulloch et al. (2025)

mapping between behavioral and physiological indicators and the METUX. Since the pupil baseline remains relatively stable and is associated with well-being, this indicator does not include an arrow.

Overall, in this study, oculomotor organization indicators that are more directly tied to the mechanisms of training behavior (e.g., fixation distribution/fixation stability and saccadic organization patterns) exhibited larger magnitudes of difference. This suggests that the training conditions were not only “statistically distinguishable,” but were also more likely to translate into perceptible changes in training behavior during real use—for instance, more stable tracking and fewer frequent switching scan-paths—thereby supporting a more consistent pattern of attentional engagement and training execution. By contrast, although physiological measures such as pupil diameter and EDA reached significance or showed trends in some comparisons, their overall effect magnitudes were more susceptible to contextual factors, including task luminance, viewing distance, and physiological arousal. Accordingly, in practical interpretation, these measures are better treated as supportive evidence that “attention/arousal has been mobilized,” rather than as standalone, decisive grounds for concluding increased motivation or engagement.

Based on the above effect-size stratification, the practical implication of this work is that subsequent system iteration and evaluation should prioritize optimizing interaction mechanisms that “facilitate stable fixations and well-structured oculomotor organization,” and should treat these as key performance indicators. Meanwhile, subjective evaluation outcomes (self-reported increases in satisfaction and alleviation of visual fatigue) should be integrated to determine whether these objective differences carry user-perceivable, practical value. With respect to potential health applications, our findings indicate that an entertaining, actively controlled training format may, in the short term, improve user experience and reduce subjective discomfort. However, given the short-duration, laboratory-based, small-sample design, the present evidence should be regarded as a pre-application proof of concept; its long-term effects and generalizability require further examination over longer periods, in naturalistic settings, and with larger samples.

6.1. Passive eye-tracking cursor visual training without device load (control group)

In passive eye-tracking training without wearable device load, participants in the control group exhibited a tendency toward concentrated and focused eye-movement trajectories (Figure 11). The MVD also demonstrated a relative advantage over the mean horizontal distance (MHD) (Figure 10). These observations suggest that basic passive eye-tracking tasks can guide participants’ gaze toward greater visual focus, thereby achieving a measurable visual training effect.

Previous studies have supported a similar perspective. Ilona Heldal et al. demonstrated the potential of integrating computer games into visual rehabilitation programs (Heldal et al., 2021), while Qasim Ali et al. employed eye-tracking combined with game-based tasks for visual training and verified its feasibility (Ali et al., 2022). Compared with traditional contact-based training approaches, screen-based eye-tracking, as a form of device-free interaction, can reduce interference from natural body movements, operational discomfort, and issues related to the bulkiness of physical equipment (Klaib et al., 2021; Liao et al., 2023).

From the perspective of the METUX model, this form of passive training shows promise at the adoption level. Owing to its simple task structure, low cognitive and physical load, and lack of reliance on specialized equipment, participants are more likely to accept and engage with the training. This, in

turn, establishes a foundation for the subsequent introduction of more active or gamified training modalities.

6.2. Proactive eye-controlled cursor visual training (Experimental Group A)

This study utilizes the METUX model to design active visual training, assessing its effects through the cross-analysis of both behavioral and physiological measures. Behavioral indicators, such as pupil size and blink frequency, are linked to physiological states such as neural load (Calancie et al., 2023; Szabadi, 2018). As illustrated in Figure 15, the cross-analysis allows for the mutual validation of results, a key feature of the METUX model's multidimensional evaluation framework.

The behavioral data show that, under the condition of active cursor control, the average pupil diameter in Experimental Group A is smaller than that in the passive tracking group, though the statistical difference is not significant. Considering the task context of near-distance fixation in this study, pupil constriction when focusing on near objects is a normal physiological response, which is part of the near reflex triad (Ghouschi et al., 2021). This finding could be linked to the increased fixation distance and the larger movement vertical distance (MVD) in Experimental Group A, with MVD significantly greater than in the control group. A key observation is that the pupil trajectory and heatmap in Experimental Group A are more concentrated (Figure 11b,e), and the frequency of saccades is higher than in the control group. This suggests an exploratory correlation with increased task engagement due to the active control design. Unlike passive training, participants must actively control eye movements to match the cursor's movement. This active engagement likely enhances attention, which is reflected in the concentrated pattern of eye movement trajectories (McColeman & Blair, 2014).

On the physiological front, the average SC in Experimental Group A is higher than in the control group, consistent with the novel experience introduced by the active control design. It is important to note that pupil dilation and increased SC are indirect indicators of physiological arousal and should not be directly interpreted as an increase in motivation or engagement. These measures can, however,

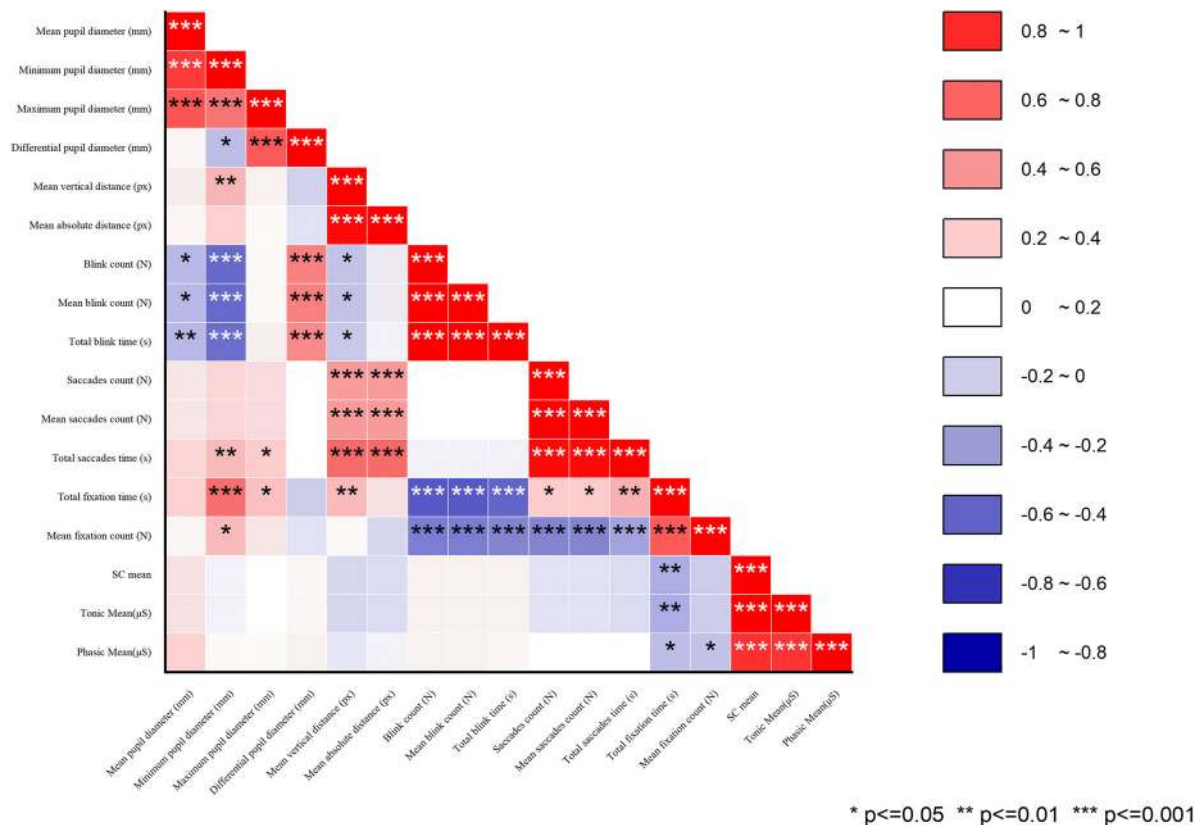


Figure 15. Correlation matrix diagram (correlation analysis of various behavioral and physiological indicators).

provide supportive evidence for the effectiveness of the active training design, while accounting for individual differences and task-specific influences.

From the perspective of interface innovation in the METUX model, the intuitive eye movement control interface developed in this study (which is lightweight, provides immediate feedback, and imposes no equipment burden) serves as the core design element that facilitates the observed behavioral and physiological effects. In contrast to traditional visual training methods that rely on complex hardware, this streamlined interface lowers the learning barrier and enhances the participants' sense of control. This approach aligns well with the METUX model's logic of "interface controllability → behavioral autonomy → motivation enhancement," representing a significant innovation that differentiates this study from conventional passive training methods.

6.3. Eye-tracking ball visual training in engaging scenarios (Experimental Group B)

This section presents the innovative design of "progressive training combined with psychological care," investigating the cumulative effects of gamified task scenarios on motivation and engagement. It also further examines the practical application of the METUX model in multi-level design settings.

In terms of behavioral metrics, Experimental Group B exhibited a smaller average pupil diameter compared to Group A. This could be attributed to the brightness and spatial complexity of the ball scene. Brighter and more intricate spatial arrangements result in pupil constriction, which sharpens the focus on the ball (Vilotijević & Mathôt, 2024). However, the primary advantage lies in the enhanced focus in the pupil trajectory (with MHD smaller in Group B than in Group A), and notably, the average fixation time was significantly longer, while blink frequency was lower. This trend may suggest a link between the gamified task design and motivational outcomes. The contextualized task of "scoring a football goal" provided clear, goal-oriented feedback, which is more effective than a simple cursor control task in sustaining attention and reducing distractions. This translated into longer fixation durations and more stable movement trajectories (Shao et al., 2021).

Additionally, the frequency of saccades was lower in Group B compared to Group A, likely related to the smooth tracking demands in ball sports. This suggests that the fun, gamified scenario led participants to adopt an attention allocation pattern better suited to the task, further supporting the motivational value of the contextualized design. Participants relied more on steady pupil size variations to control the ball, as opposed to frequent eye movements that shifted focus. This behavior may be linked to interest-driven attention stability, and the potential increase in interest could be related to dopamine release (Matsumoto & Hikosaka, 2009; Oh et al., 2012). However, it is important to note that this remains a theoretical hypothesis and cannot be confirmed as a causal relationship based on the available data.

On the physiological front, Experimental Group B had higher average values for SC and SCL compared to Group A, aligning with the trend of emotional activation triggered by the gamified design (Bari et al., 2023; Yu et al., 2024). However, it should be emphasized that these physiological changes are indirectly associated with motivation and engagement, and no causal inferences can be drawn. Subjective evaluations suggest that the gamified design could enhance physiological activation and increase task engagement.

From the perspective of the METUX model's multi-level design logic, the innovative value of Experimental Group B is evident in its progressive optimization over Group A:

1. Task-level innovation: Visual training was embedded in the "football goal" gamified task, using clear goal setting and immediate feedback to reinforce the METUX model's transition from "task-level clarity" to "behavior-level engagement."
2. Behavior-level enhancement: More stable fixation patterns and higher trajectory focus are direct reflections of the gamified task design at the behavioral level, improving behavioral consistency and task involvement compared to the simple cursor control in Group A.
3. Life-level implications: The positive trend observed in subjective assessments suggests that the gamified design can increase training acceptance, providing clues for future long-term real-life training. However, due to the small sample size (N=20), laboratory setting, and short task

duration, the long-term effects on happiness cannot be confirmed. Thus, this only provides exploratory support for the innovative concept of integrating psychological care into training design.

A comparison of Experimental Groups A and B clearly illustrates the progressive logic of the design: from “no device burden with active control” (addressing the dependency on traditional training equipment) to “gamified scenario incentives” (addressing the challenge of sustaining active training). This layered approach aligns with the METUX model’s goal of integrating motivation, engagement, and happiness. Both groups exhibited exploratory trends supporting the design’s effectiveness (behavioral focus and physiological activation). However, it should be emphasized that all conclusions are exploratory and indicative, not confirmatory. Given the limitations of sample size, laboratory settings, and short task durations, no causal relationship between the design intervention and motivation/engagement can be confirmed.

It is important to note that physiological indicators such as pupil size, SC, and SCL are indirectly related to the METUX constructs (motivation, engagement, and happiness). These interpretations are specific to this study’s task design (close-range visual training, gamified tasks) and cannot be generalized. The current findings are based on the consistency between design logic and data trends, rather than direct task-specific validation. Future research should involve larger sample sizes, longer training periods, and real-life scenarios to further assess the validity of these indicators.

6.4. System relief effects and advantages analysis

6.4.1. Visual fatigue relief effects

As analyzed in the results section, all three groups exhibited a reduction in visual fatigue symptoms following training, though the effectiveness varied significantly across the different training modes. Group B demonstrated the most notable improvement, particularly with regard to symptoms such as dry eyes, blurred vision, and general eye discomfort. This suggests that the gamified design, by offering more engaging and varied interactive experiences, not only enhanced user involvement but also likely increased emotional activation, which in turn contributed to the alleviation of visual fatigue. The situational design of the gamified training further boosted user interest and task engagement, positively impacting the reduction of visual fatigue symptoms.

In contrast, although Group A did not achieve the same degree of improvement as the gamified training, it still showed notable relief, especially for dry eyes and blurred vision. Active training, by requiring users to interactively control interface elements, strengthened the overall engagement and focus, which may have led to more effective alleviation of visual fatigue. Notably, active participation in the training provided users with a sense of control and accomplishment, which helped reduce eye discomfort through a self-regulatory process.

In comparison, the control group, while showing some improvement, had a significantly weaker response than both the active and gamified training groups. The passive training mode, lacking both user control and a rich contextual design, resulted in lower engagement, thereby offering weaker relief from visual fatigue. Nevertheless, this outcome still indicates that passive training can reduce visual fatigue to some extent, confirming the overall effectiveness of the system design.

Furthermore, after completing the training, participants from all groups reported feeling more relaxed in their eyes, with an improved comfort level when using electronic devices. A key advantage of this system is that it requires no additional hardware—unlike traditional training methods that involve wearing eye trackers or virtual reality headsets. This absence of extra equipment eliminates the discomfort associated with such devices, enhancing both user comfort and system acceptability. Participants consistently reported that their dry eyes and discomfort were alleviated, and they could use electronic devices for extended periods while maintaining a high level of comfort.

6.4.2. Advantages

There are several eye-tracking and visual training systems currently available on the market, such as visual rehabilitation games based on eye tracking and apps for eye relaxation training (Chhimpa et al.,

2025; Erdag et al., 2025). Most of these systems require additional hardware (such as specialized eye trackers or virtual reality devices), making the equipment uncomfortable to wear or limiting user mobility. In contrast, the METUX-based eye-tracking visual training system proposed in this study offers the advantage of requiring no additional equipment. Users can perform the training using a standard display and basic eye-tracking technology, avoiding the discomfort and bulkiness of additional devices, which improves the system's comfort and acceptability.

Moreover, traditional training systems often lack engaging elements, which can lead to lower user involvement. In contrast, the system in this study incorporates gamified elements and active control, significantly boosting user motivation and engagement, and providing a marked improvement in user experience, particularly in reducing visual fatigue. Participants generally reported reduced eye discomfort and fatigue after using the system, along with increased comfort during screen use.

The reduction in visual fatigue not only improved eye comfort but also enhanced the participants' overall sense of well-being. In this study, the combination of gamified design and active control not only helped alleviate eye discomfort but also improved participants' moods. Survey results indicated that participants felt more relaxed and content following the visual training. They also reported feeling more motivated and engaged, with the enjoyable nature of the training significantly enhancing their emotional activation levels. Some participants mentioned that the improvement in eye comfort and emotional state made their subsequent use of screens more pleasant and relaxing. This indicates that the system not only alleviates visual fatigue physiologically but also enhances users' emotional well-being.

Overall, the significant effects of the gamified training model suggest that integrating active participation, interest-driven design, and task engagement holds great potential for reducing visual fatigue. Introducing challenging and enjoyable scenarios can increase user involvement, thereby enhancing training effectiveness. Future research could explore additional ways to combine various gamified designs with active training to further optimize visual fatigue relief systems.

6.5. Limitations and future work

Although this study provides valuable preliminary, proof-of-concept evidence for interaction design in visual training grounded in the METUX model, several limitations remain and must be carefully considered when interpreting and generalizing the findings. First, the sample size was small ($n = 20$) and participants were recruited from a relatively narrow source, which may reduce statistical power and limit external validity and generalizability. Future studies should replicate the investigation with larger and more diverse populations to strengthen the robustness of the conclusions. Second, the gender distribution was notably imbalanced (15 males and 5 females). Prior research suggests that gender differences may influence behavioral responses related to motivation and engagement, as well as certain physiological indicators; therefore, the present findings may be affected by potential sampling bias. Subsequent work should adopt a more balanced gender composition and, where feasible, conduct stratified analyses or include gender as a covariate to mitigate bias and improve generalizability. Third, the task duration was brief and the study employed a single-session design, which limits the ability to capture sustained changes in motivation and engagement during real-world training and prevents evaluation of long-term use after novelty effects diminish. Accordingly, longitudinal studies over extended periods are needed to examine the system's impact on sustained engagement, adherence, and the formation of behavioral habits. Fourth, the study was conducted in a controlled laboratory setting. While this facilitates variable control and enhances internal validity, the controlled context may produce participant behaviors that differ from everyday use, thereby constraining ecological validity. Future research should incorporate long-term follow-up in more naturalistic settings (e.g., home or routine training environments) to validate usability and effect stability in real-world contexts and to further assess long-term impacts on eye-health-related outcomes.

In summary, given constraints related to sample size and composition, experimental duration, and study context, the findings are best regarded as exploratory and proof-of-concept evidence for METUX-informed interaction design in visual training—useful for generating design hypotheses and research directions rather than serving as definitive design principles for broad dissemination. The core

contribution of this work lies in its initial validation of the feasibility of METUX-based, device-free proactive training and gamified progressive training, providing a clear theoretical and methodological foundation for future large-sample, long-term, and naturalistic studies.

6.6. Implications for future design

This study offers several directions for the system's subsequent development. To enable flexible use across scenarios, a modular architecture and cross-platform technologies could allow core algorithms and business logic to run seamlessly on servers or clients. Future extensions may include mobile support (e.g., iOS/Android apps or PWAs), offline data synchronization, and adaptation to low-network conditions, enhancing adoption and long-term engagement. Non-critical modules could be open-sourced to foster community participation and open innovation, while critical algorithms and user privacy data must remain strictly protected to ensure user trust and adherence to ethical standards. The system's core "eye-controlled ball" interaction can be further optimized to provide smoother tracking, richer feedback, and personalized difficulty adjustments, thereby enhancing user motivation and training efficacy. Beyond alleviating digital eye strain, the system could be expanded for ophthalmic rehabilitation interventions such as amblyopia, accommodation disorders, and strabismus, demonstrating potential contributions to public health. Future work will also focus on personalized recommendation mechanisms, edge-side resource optimization, and the construction of an open ecosystem to support sustainable long-term development across multi-platform, multi-user, and multi-task scenarios.

7. Conclusion

This study, grounded in the METUX model, introduces a visual training strategy that requires no additional device load, designed to enhance motivation, engagement, and well-being through eye relaxation to alleviate digital eye fatigue. Initially, the feasibility of a device-free design was validated, followed by an analysis of experimental data on active and gamified visual training. Through a three-stage progressive experiment, the training transitioned from passive to active, with gamified elements incorporated into the active phase. The results indicate that active control and gamified design can enhance participant engagement in visual training, thereby improving motivation, engagement, training efficiency, and effectiveness, while also fostering well-being.

This design aims to empower participants with greater autonomous control over their visual training through technological support, aligning the training process with personal goals and values, addressing psychological needs, and promoting a human-centered approach. This study represents the first application of innovative HCI methods to enhance intrinsic motivation, facilitating more effective and lasting behavioral changes. It enables participants to enjoy the training process while improving vision and enhancing well-being. We contend that design can not only create enjoyable experiences but also reflect individual meaning and goals, guide behavioral intentions, and motivate individuals to engage in activities that have personal and societal significance.

Overall, the findings suggest that gamified active visual training can significantly improve three core expectations of user experience: motivation, engagement, and well-being. These results provide new insights for the field of Positive Computing, support interdisciplinary exploration aimed at enhancing human well-being, and offer fresh perspectives on digital experience design. For certain indicators that did not show significant differences, this may be attributed to the limited sample size or potential errors due to posture adjustments as participants adapted to the active training. Future work can build on these findings to improve the design. Currently, a simplified model suitable for real-time applications has been developed. Plans involve designing new experiments, utilizing publicly available datasets, focusing on active visual training, and expanding the participant pool to strengthen the robustness and generalizability of the solution.

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Authors' contributions

CRedit: **Xiaoying Li**: Funding acquisition, Resources, Supervision; **Haoyu Zhang**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Guangran Li**: Supervision.

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Data availability statement

The datasets generated or analyzed during this study are available from the corresponding author on reasonable request.

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